The ten-point symmetric factor

Human Compression project: a component of the L019 recovery

All actions below are actual permutation actions. A marking is *fractured* when every maximal omission is intransitive. An endpoint anchor is moved by the first mark and fixed by all later marks, or by the reversed version of that condition.

Lemma 1. *In a fractured transitive marking, each mark of every prefix is essential to that prefix's orbit partition.*

Proof. The orbits of a generated permutation group are the equivalence classes generated by the pairs x, x^s for its generators s. If a prefix and one of its omissions have the same orbit partition, adjoining the same next generator to both gives the same partition again. Inductively they still have the same partition after all remaining generators have been adjoined. The full marking is transitive, so the corresponding full omission would be transitive too, contradicting fracture. □

Theorem 2. *Let $(t_0, \dots, t_5)$ be a fractured natural S_{10} string C-marking with endpoint anchor a. Suppose that, on the same ten-point domain, it satisfies*

$$P_j \cap \mathrm{Alt}(X_{j-1}) \le P_{j-1} \quad (1 \le j \le 6), \qquad P_j = \langle t_0, \dots, t_{j-1} \rangle, \quad X_j = a^{P_j}. \tag{1}$$

Then t_0 is a pure transposition.

Proof. We give a complete finite exclusion of a nonpure t_0. Conjugate the anchor to point 1. All involutions of S_{10} are obtained by taking the five conjugacy classes represented by products of one, two, three, four or five disjoint transpositions. There are 9495 nonidentity involutions, of which 2619 fix 1. At the first position we require movement of 1 and support larger than two. At each later position we try every involution fixing 1 which commutes with every previous mark except its immediate predecessor, does not commute with that predecessor, and does not belong to the group of the previous prefix.

Adjacent noncommutation is necessary here. In a transitive anchored string action, let H be generated by the connected component of the noncommutation diagram containing the first mark. Every other component centralizes H and fixes 1. The full orbit of 1 therefore equals its H-orbit, so H is transitive. A permutation centralizing a transitive action and fixing one point fixes all points. There can consequently be no nonidentity mark in another component. Membership in the previous prefix is excluded by IP.

Only conjugacy repetitions are removed. Initially conjugate by $(S_{10})_1$. After marks have been chosen, conjugate the next candidate only by the subgroup fixing 1 and centralizing every chosen mark. This subgroup preserves all stated conditions. Each of its orbits contributes one representative, so induction on the prefix length proves that every possible marking remains represented. No arbitrary graph identification or restriction quotient is involved.

For a prefix group H of length l, the necessary order condition is $[S_{10} : H] \ge 2^{6-l}$: each remaining independent generator at least doubles its preceding subgroup. Every trial prefix is also tested by Lemma 1 and by (1). The latter is computed as an actual intersection with the group of even permutations supported on the previous anchor orbit. For orbits of size at most two that group is trivial.

The interval intersection tests are also necessary consequences of the original IP. When appending a mark at position l, put

$$H_i = \langle t_i, \dots, t_{l-1} \rangle, \qquad J_i = \langle t_i, \dots, t_l \rangle.$$

For every $0 \le i \le l-2$ require

$$H_i \cap J_{i+1} = H_{i+1}. \tag{2}$$

The old groups H_i are retained from the preceding step. These tests avoid repeatedly constructing every subset group at every node. For the exclusion, only their necessity is needed. At every terminal node the implementation nevertheless checks all pairs of all 64 generated subset groups directly. Thus its reported terminal markings also satisfy the full IP.

The complete counts of visited prefixes, including the empty one, are

$$1, 4, 197, 1775, 645, 208, 18.$$

All four nonpure first-mark types are exhausted. The orbit test rejects 3949 candidates, (2) rejects 1336, and (1) rejects 1402 further candidates, in that order. None of the eighteen terminal groups has order $10! = 3628800$. Since all are subgroups of the given faithful S_{10}, none is the required natural group. The literal source program, full terminal receipt, terminal orders and orbit sizes are retained. Every rejection was justified by a necessary condition, and every conjugacy reduction was exhaustive. This proves the finite exclusion and hence the theorem. □

Theorem 3. *Consider the actual critical one-point configuration retained in the L019 recovery: a fractured natural symmetric factor P of even degree m and rank $m/2+1$, and an even six-point rank-three A_5 factor E, supported on X, Y with $X \cap Y = \{a\}$. Their labels are consecutive disjoint intervals, and a is an endpoint anchor in each. Assume the whole marking is a string C-marking. If $m = 10$, the whole marking has an internal pure perfect split. Together with the acquired smaller cases, only $m \ge 12$ remains in this critical configuration.*

Proof. Orient P with its anchor first. The acquired odd-attachment theorem supplies (1) from the IP of the whole marking and the actual external transitive group E. It requires no odd element in E for this even-kernel conclusion. Theorem 2 gives $t_0 = (a\ c)$.

The tail of P fixes a and is transitive on $X \setminus \{a\}$. Indeed, if it had more than one orbit there, adjoining the sole edge ac could join a to only one of them, contrary to transitivity of P. In the whole marking, omission of t_0 therefore leaves precisely Y and $X \setminus \{a\}$ as its orbits. The two label intervals are supported and transitive on their respective sets, and the omitted mark is pure. This is an internal pure perfect split. The acquired pure-split criterion supplies its valid rank and degree extension. The separately reviewed small-symmetric component settles $m \le 8$; no larger degree is being inferred from the present finite calculation. □

Attribution and limits. The principal source is Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2. The critical reduction, supported even-kernel theorem and precise split criterion are acquired with their attributions and review chains. The present contribution is the orbit-partition pruning argument and this complete bounded ten-point exclusion. Permutation-group operations use GAP; the conjugacy and branching completeness argument is given above. An independent small-group check compares the interval tests with all subset intersections, and a retained pure-first S_{10} example is checked as a positive control.

Two earlier degree-ten searches stopped at their declared caps without a certificate. Their receipts remain unchanged. A first attempt at the present program stopped before search because its global variable name was reserved in GAP; the corrected complete run and the precise change are recorded. These earlier executions carry no exclusion credit. The general critical case, contraction, the stability correspondence under isomorphism and duality, and the exact base counts remain pending. No whole-original, new question, Lean or compression-measurement credit is asserted.